

جامعة بيروت العربية
BEIRUT ARAB UNIVERSITY

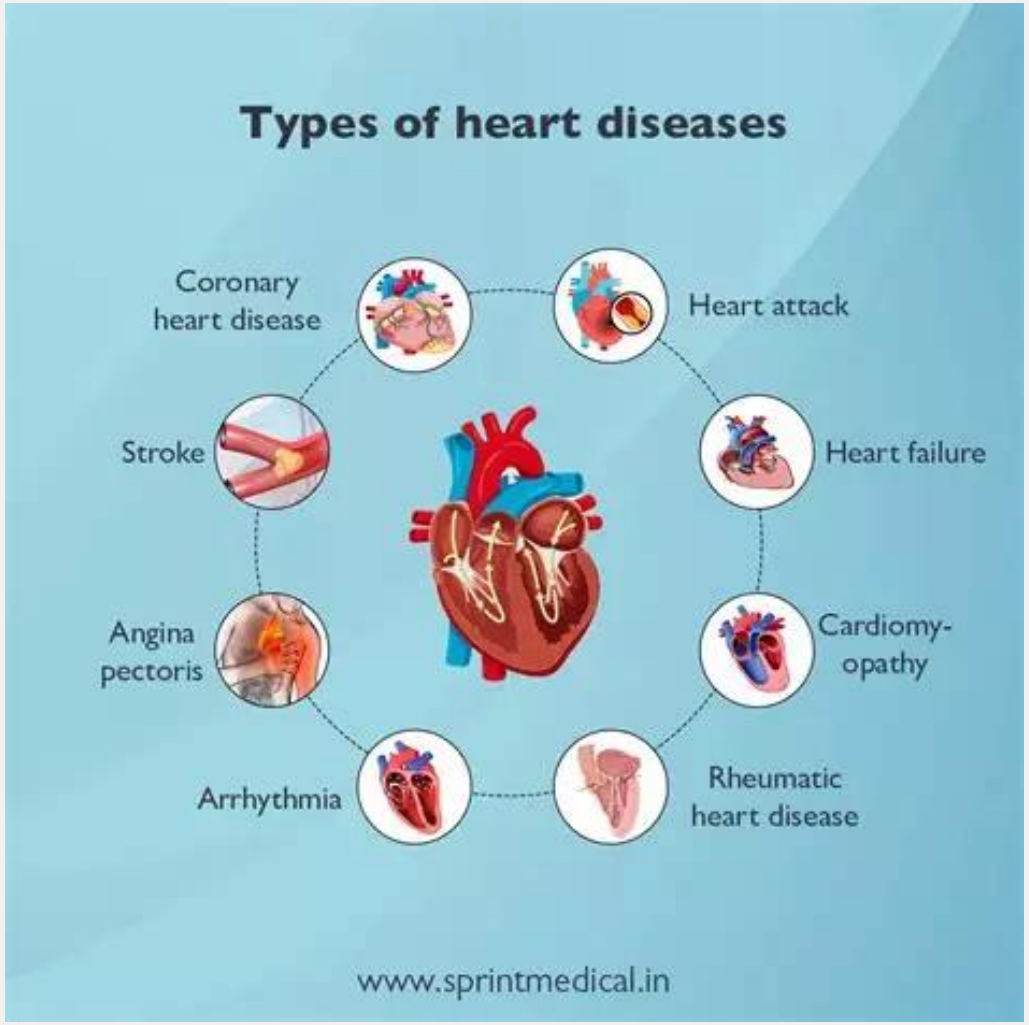
Course: CMPS452

Heart Disease Prediction Using Random Forest

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Problem Statement

- Heart disease is a leading cause of death
- Early detection is difficult
- Traditional diagnosis takes time
- Need for fast prediction system



Motivation

- Early prediction can save lives
- Machine learning can help doctors
- Reduce human errors
- Faster decision making



Project Objective

1. Predict heart disease (0 / 1)
2. Use Random Forest model
3. Compare with Decision Tree
4. Achieve high accuracy

Dataset

Heart Disease Dataset

- 573 records
- 14 features

Examples

- Age
- Cholesterol
- Blood Pressure

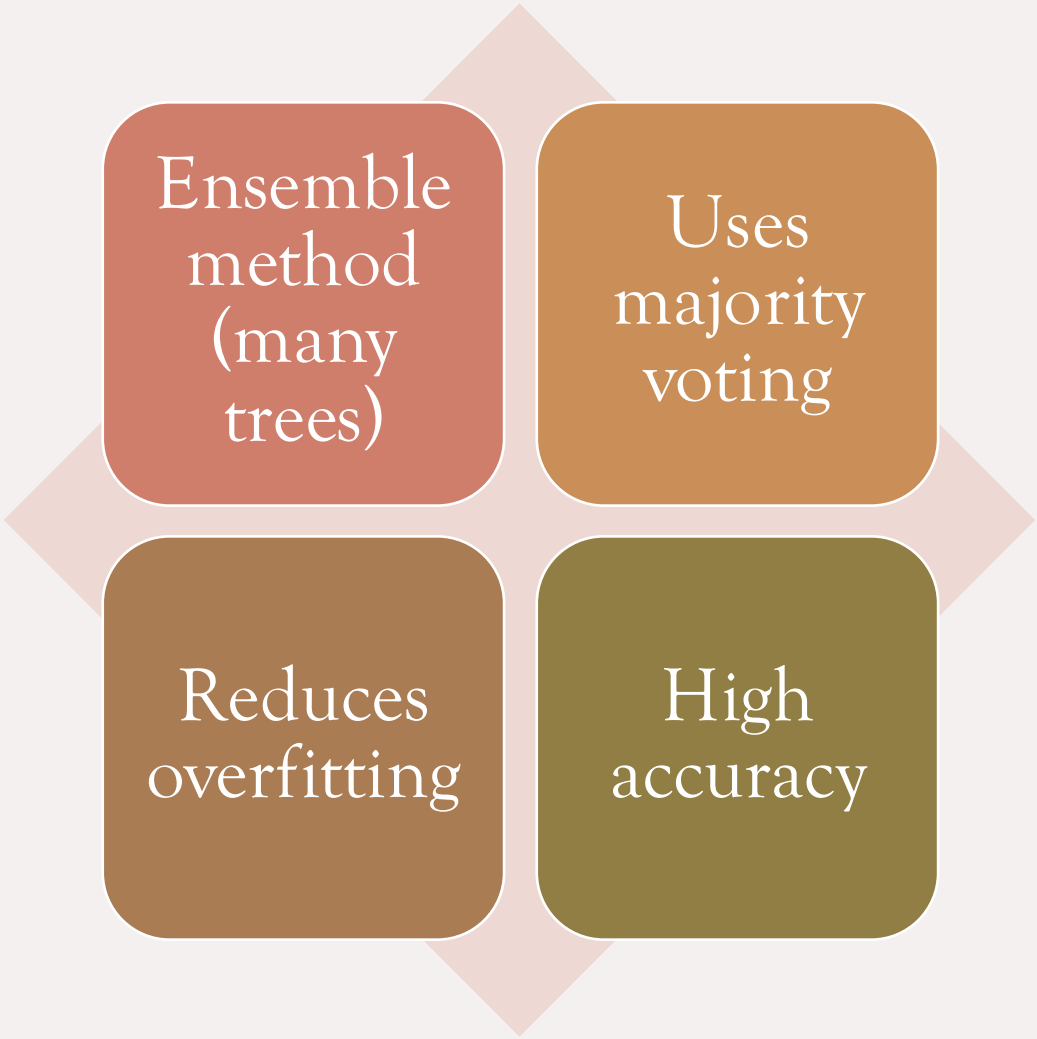
Target

- 0 = No disease
- 1 = disease

Methodolgy

Supervised Learning	Classification problem	Train/Test split (80/20)
Models used:	Random Forest	Decision Tree

Random Forest



Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score

Results

Model	Accuracy	Precision	Recall	F1
Random Forest	0.973	0.981	0.962	0.971
Decision Tree	0.956	0.980	0.925	0.952

- Random Forest performs better

Discussion

Random Forest is better than Decision Tree

Decision Tree overfits

Dataset size is small

Results are good overall

Conclusion



MACHINE LEARNING CAN PREDICT HEART DISEASE



RANDOM FOREST IS THE BETTER MODEL



GOOD ACCURACY ACHIEVED



USEFUL FOR HEALTHCARE

Future Work



USE LARGER DATASET



TRY OTHER ALGORITHMS



IMPROVE FEATURES



BUILD REAL APPLICATION

Thank you ☺